NISHANTH SUNDARAN

Robot Learning | Imitation & Reinforcement Learning, Real Robots, MuJoCo / MJX

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Work permit: Germany, § 20 AufenthG (post-study residence permit); willing to travel to deployment sites



PROFILE

Early-career robot learning engineer (M.Eng. Artificial Intelligence, TH Deggendorf, 2026). I train neural policies for humanoids, from human motion-capture data through GPU training in JAX / MJX to evaluation against measured human references, and I care most about whether the numbers are true: I have caught reward hacking, simulator mismatches and evaluation bugs that would otherwise have produced believable but wrong results. On the hardware side, my master's thesis ran my own ROS 2 code (Python and C++) on a real UR3e with force/torque sensing and a depth camera, validated over 120+ logged runs. I move comfortably between research code and physical robots, use AI tools daily while checking their output against measurements, and am happy to travel to customer sites.

ROBOT LEARNING PROJECTS

G1 Motion Imitation: Learning a Human Gait from Motion Capture (JAX / MJX, PPO)

Open source, since 08/2026: github.com/NishanthSundaran/g1-motion-imitation

- **Data to policy pipeline:** DeepMimic-style imitation learning of LAFAN1 human mocap on a Unitree G1 (LocoMuJoCo, mimic setup adapted from the H1 to the G1), trained with PPO in MJX at approx. 30,000 steps/s on a 6 GB laptop GPU (9x the CPU baseline); about 5.8B steps over 13 resumed training cycles.
- **Evaluation design:** every cycle judged by the same MJX evaluation script against the same human clip on speed, cadence, pelvis bob, stride, knee flexion, foot clearance and survival; fixed three measurement errors (CPU vs. MJX mismatch, auto-reset frames, wrong reference window) that would each have produced believable but false numbers.
- **Failure diagnosis:** found the policy exploiting a per-step reward by surviving at 2.6x human speed with scrambling steps; closed the loophole with a root-deviation termination and re-weighted the reward toward joint tracking. Showed PD position control reached in 5M steps what torque control needed 300M for.
- **Result:** four of six gait metrics within 7% of the human reference (speed and cadence within 2%); honest limitation: the robot still falls after about 1.5 s on average, a compute limit since survival rose steadily with steps. Contributed four undocumented framework fixes upstream.

One Brain, Many Bodies: Whole-Body Policies and Hierarchical Autonomy for Humanoids (MuJoCo)

Open source, since 08/2026: github.com/NishanthSundaran/one-brain-many-bodies

- **Whole-body policies:** PPO stance policy (14 CPU workers) that holds the pelvis to 1.4 cm while the arms reach; a learned reach policy rewarded on tracking a human minimum-jerk trajectory was more accurate (0.024 m, 3/3) than one rewarded on final distance (0.046 m) or deterministic IK (0.075 m, 0/3).
- **Benchmarking before deciding:** compared learned whole-body control with IK on identical targets in two scenes; the policy won in open space (0.022 vs. 0.408 m) but lost at the work bench (0.516 vs. 0.082 m), so grasping uses IK. A unified locomotion and manipulation policy (9M steps) was measured worse than the two specialists and documented as a negative result.
- **Hierarchical autonomy:** a local vision-language model (Qwen2.5-VL-3B) plans at about 1 Hz through schema-constrained tool calls while the controllers run at about 500 Hz; the same stack drives Unitree G1 and H1. 28 automated tests.

Embodied Agent: VLM-grounded Pick-and-Place (ROS 2, MoveIt 2, Gazebo)

Portfolio project, 04/2026: github.com/NishanthSundaran/embodied_agent

- A VLM grounds objects in camera images, depth back-projection gives 3D targets, and an LLM decomposes language commands into verified pick and place steps on a simulated UR3e with RealSense D435i.

RESEARCH AND WORK EXPERIENCE

09/2025 – 06/2026 **Master's Thesis Researcher, real-robot human-robot interaction (UR3e)**
Technische Hochschule Deggendorf (THD), Technologie Campus Cham, Germany
Thesis: Human-Robot Interaction with Force Feedback in Collaborative Tasks; defended 14.06.2026. Code: github.com/NishanthSundaran/Human-Robot-Interaction-with-Force-Feedback-in-Collaborative-Tasks

- Ran my own ROS 2 code (Python and C++) on a physical UR3e with force/torque sensor, Intel RealSense D435i and Robotiq 2F-140, and debugged what changed between simulation and hardware (e.g. trajectories tripping joint acceleration limits near the reach boundary).
- Multimodal sensing: YOLOv8 segmentation removes the person from the depth point cloud for collision-free planning; a force/torque classifier separates deliberate guiding from short nudges; both are fused to scale robot speed around people.
- 500 Hz admittance control (also released as an open-source C++ ros2_control controller) and a 13-state task state machine for assistive lift-and-place; 120+ logged runs in Gazebo and on the real robot.

05/2026 **Robot Data Collection Specialist (freelance via Zenjob, 2 days)**
AlphaIgnis GmbH, Munich, Germany

- Collected bimanual robot demonstrations by teleoperating two Franka arms with joysticks on a single coordinated task, producing training data for robot learning.

12/2023 – 11/2024 **Machine Learning Intern**
Energy Future World, London, United Kingdom (remote)

- Trained and benchmarked computer vision and ML models on real-world aerial imagery; 85% detection accuracy.

03/2022 – 12/2022 **Process Associate**
Amazon Development Centre, Bangalore, India

- Data analysis and auditing for Amazon retail operations.

05/2019 – 11/2021 **Project Engineer**
Power And Instrumentations Ltd, Gujarat, India

- Delivered high-voltage power projects on site: commissioning, contractor coordination and field documentation.

TECHNICAL SKILLS

Robot Learning	Imitation learning from human motion capture (DeepMimic-style), reinforcement learning (PPO), reward design, whole-body and locomotion policies, evaluation and ablation design, failure analysis
Frameworks	JAX / MJX, PyTorch, MuJoCo, LocoMuJoCo, Gazebo, ROS 2 (Humble), MoveIt 2, ros2_control
Models	Vision-language models as planners (Qwen2.5-VL via Ollama, Claude), schema-constrained outputs, YOLOv8 detection and segmentation
Hardware	Universal Robots UR3e, Franka arms (bimanual teleoperation), Robotiq 2F-140, force/torque sensing, Intel RealSense depth camera, hand-eye calibration; Unitree G1 / H1 (simulation)
Software	Python, C++, Git, Linux, pytest, CMake

EDUCATION

10/2022 – 06/2026 **M.Eng. Artificial Intelligence for Smart Sensors and Actuators**, Technische Hochschule Deggendorf, Germany. Focus: Machine Learning, Deep Learning, Autonomous Systems, Robotics, Computer Vision.
B.Eng. Electrical and Electronics Engineering, University of Calicut, Kerala, India.

LANGUAGES

English	Professional fluency, written and spoken (C1)
German	A2 (elementary), actively learning toward B1

CERTIFICATIONS

09/2026 **Machine Learning with Python**, IBM (Coursera). Supervised and unsupervised learning, classification, regression, clustering, model evaluation, Scikit-learn. Verify credential

08/2026 **AI Agents and Agentic AI with Python & Generative AI**, Vanderbilt University (Coursera). Tool use, function calling, LLM application development, RAG, prompt engineering. Verify credential